

Difference finite

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$$\partial_z^2 u = \left[c_z^2 \partial_z^2 + c_t^2 \nabla_\perp^2 \right] u, \quad K_z(z) = \frac{n\pi}{2L(z)}, \quad L(z) = L_0 - \frac{h}{2} \tanh\left(\frac{z-R_0}{\varepsilon}\right)$$

$$u(x, z, \vartheta, t) = \cos\left(K_z(z)x\right) T(x, \vartheta) e^{i\omega t}$$

$$-\omega^2 T = -c_z^2 K_z^2 T + c_t^2 \nabla_\perp^2 T$$

$$\left[-\left(\frac{\omega}{c_t}\right)^2 + \left(\frac{c_z}{c_t}\right)^2 K_z^2 \right] T = \nabla_\perp^2 T$$

$$c_t^2 K_z^2(z) + c_z^2 K_z^2(z) = \omega^2$$

$$\Rightarrow -K_z^2(z) = -\left(\frac{\omega}{c_t}\right)^2 + \left(\frac{c_z}{c_t}\right)^2 K_z^2(z)$$

$$-K_z^2(z) T = \left[\frac{1}{z} \partial_z (z \partial_z) + \frac{1}{x^2} \partial_\vartheta^2 \right] T$$

$$T(x, \vartheta) = f(x) g(\vartheta)$$

$$K_z^2 f g = \frac{g}{z} (f' + z f'') + \frac{f}{x^2} \partial_\vartheta^2 g$$

$$-K_z^2 - \frac{1}{z} \left(f'_f + z f''_f \right) = \frac{\partial^2}{g}$$

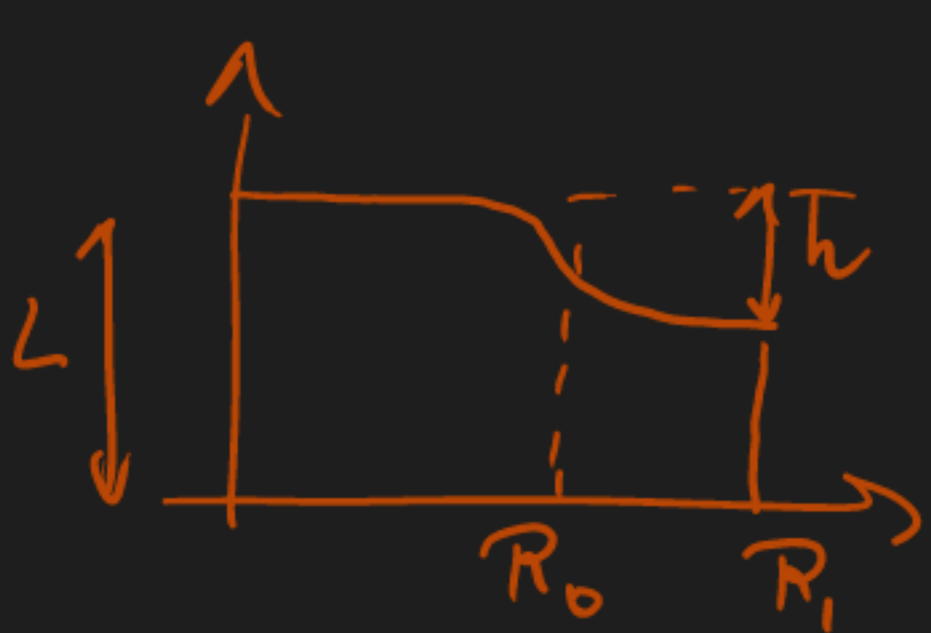
$$K_z^2 + \frac{1}{z} \left(\frac{f'}{f} + z \frac{f''}{f} \right) = -\frac{g}{f} = \mu^2$$

Equazione separata

$$K_z^2 - \mu^2 + \frac{1}{z} \left(\frac{f'}{f} + z \frac{f''}{f} \right) = 0$$

$$(K_z^2 z^2 - \mu^2) f + z f' + z^2 f'' = 0$$

$$\left[\left(\frac{\omega}{c_t}\right)^2 - \left(\frac{c_z}{c_t}\right)^2 K_z^2 - \frac{\mu^2}{z^2} \right] + \frac{1}{z} f' + f'' = 0$$

$$\frac{d^2 f}{dz^2} + \frac{1}{z} \frac{df}{dz} - \underbrace{\left[\left(\frac{c_z}{c_t}\right)^2 K_z^2(z) + \frac{\mu^2}{z^2} \right]}_{V^{m,m}(z)} f = -\left(\frac{\omega}{c_t}\right)^2 f$$


$$-\frac{1}{z} \frac{d}{dz} \left(z \frac{df}{dz} \right) + V^{m,m}(z) f = \left(\frac{\omega}{c_t}\right)^2 f$$

$R_1 = \text{zorggio esterno}$
 $R_0 = \text{ " interno}$

Pseudo ordinario

$$p = \frac{z}{R_1} \Rightarrow \frac{d}{dz} = \frac{d}{dp} \frac{dp}{dz} = \frac{1}{R_1} \frac{d}{dp}$$

$$\left[\frac{1}{p R_1} \left(\frac{1}{p} \frac{d}{dp} \right) \left(p R_1 \frac{1}{R_1} \frac{d}{dp} \right) + V(p) \right] f = \left(\frac{\omega}{c_t}\right)^2 f$$

$$\left[-\Omega = \frac{\omega R_1}{c_t} \right]$$

$$\left[\frac{1}{p} \frac{d}{dp} \left(p \frac{d}{dp} \right) + V^{m,m}(p) f \right] = \Omega^2 f$$

DIFFERENCE FINITE

$$p_i = \left(i + \frac{1}{2}\right) \Delta p, \quad \Delta p = \frac{1}{N} \quad (N = \# \text{ pti. grid})$$

$$p_{i \pm \frac{1}{2}} = p_i \pm \frac{\Delta p}{2}$$

$$\left(\frac{df}{dp} \right)_{i+\frac{1}{2}} \approx \frac{f_{i+1} - f_i}{\Delta p}, \quad \left(\frac{df}{dp} \right)_{i-\frac{1}{2}} \approx \frac{f_i - f_{i-1}}{\Delta p}$$

$$\left(p \frac{df}{dp} \right)_{i+\frac{1}{2}} = p_{i+\frac{1}{2}} \frac{f_{i+1} - f_i}{\Delta p}, \quad \left(p \frac{df}{dp} \right)_{i-\frac{1}{2}} = p_{i-\frac{1}{2}} \frac{f_i - f_{i-1}}{\Delta p}$$

$$\frac{d}{dp} \left(p \frac{df}{dp} \right)_i = \frac{(p f')_{i+\frac{1}{2}} - (p f')_{i-\frac{1}{2}}}{\Delta p} = \frac{p_{i+\frac{1}{2}} (f_{i+1} - f_i) - p_{i-\frac{1}{2}} (f_i - f_{i-1})}{\Delta p^2} =$$

$$-\frac{1}{p} \frac{d}{dp} \left(p \frac{df}{dp} \right)_i = \frac{1}{(\Delta p)^2} \left[\frac{p_{i+\frac{1}{2}}}{p_i} (f_{i+1} - f_i) - \frac{p_{i-\frac{1}{2}}}{p_i} (f_i - f_{i-1}) \right]$$

$$= \underbrace{\left(-\frac{p_{i+\frac{1}{2}}}{(\Delta p)^2 p_i} \right)}_{c_i} f_{i+1} + \underbrace{\left(\frac{p_{i+\frac{1}{2}} + p_{i-\frac{1}{2}}}{(\Delta p)^2 p_i} \right)}_{b_i} f_i + \underbrace{\left(-\frac{p_{i-\frac{1}{2}}}{(\Delta p)^2 p_i} \right)}_{a_i} f_{i-1}$$

$$-\frac{1}{p} \frac{d}{dp} \left(p \frac{df}{dp} \right)_i = a_i f_{i-1} + b_i f_i + c_i f_{i+1} = \mathbb{D} \in \text{Mat}(N-1, N-1)$$

BOUNDARY CONDITIONS

STRESS-FREE
BC

$$\frac{df}{dp} \Big|_{p=1} = 0 \quad (\Rightarrow \mathcal{L}f|_{p=1} = 0) \Rightarrow \frac{f_{N+1} - f_{N-1}}{\Delta p} = 0 \Leftrightarrow f_{N+1} = f_{N-1}$$

$$\mathcal{L}_N = -\frac{1}{p_N} \frac{p_{N+\frac{1}{2}} (f_{N+1} - f_N) - p_{N-\frac{1}{2}} (f_N - f_{N-1})}{(\Delta p)^2}$$

$$\stackrel{f_{N+1} = f_{N-1}}{=} -\frac{1}{p_N} \frac{p_{N+\frac{1}{2}} (f_{N-1} - f_N) - p_{N-\frac{1}{2}} (f_N - f_{N-1})}{(\Delta p)^2}$$

$$= \frac{2}{(\Delta p)^2} \frac{(p_{N+\frac{1}{2}} - p_{N-\frac{1}{2}}) f_N}{p_N} = \frac{2}{(\Delta p)^2} (f_N - f_{N-1})$$

$$\Rightarrow \mathcal{L}_N = \frac{2}{(\Delta p)^2} (f_N - f_{N-1})$$

derivate costante in
nel punto N-esimo